

I'm a computer science PhD student at Cornell interested in building better decision making systems with optimization and machine learning methods, with applications in transportation and resource management.

## EDUCATION

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- 2021 - Present      **Ph.D. Computer Science**, Cornell University  
Advisors: Andrea Lodi, David Shmoys  
Research areas: discrete optimization, machine learning, operations research
- 2020                  **B.Sc. Honours Computer Science**, University of British Columbia

## RESEARCH

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- 2025 - Present    **Digital Twins for Transit Planning**  
Developed a digital twin for the New York bus system to simulate and compare interventions using either open data or internal data obtained through an MTA partnership. Compared four interventions on over 300 routes, finding bus spacing is often impactful than speed increases. Modeling and designing interventions with discrete optimization.
- 2024 - Present    **Optimization for Fisheries Planning, Cornell University**  
Working with fisheries experts to model and design interventions to reduce community-level income variability in Alaska using mathematical programming and machine learning methods. Initial models show the potential to reduce income variability by over 50% and highlight vocational training as a policy lever.  
*Greenstreet, L., Q. Shi, M. Grimson, F.W. Simon, S.A. Sethi, D. Shmoys, A. Lodi. Reducing Income Variability in Natural Resource Portfolios via Integer Programming. CPAIOR, 2025. doi.org/10.1007/978-3-031-95976-9<sub>2</sub>*
- 2022 - 2024      **Deep Learning for Aquaculture Mapping, Cornell University**  
Developed deep learning models to map aquaculture development from satellite imagery, using multi-spectral data, contrastive learning, and masking to handle few labeled examples and generalize to new regions. Models were used by ecologists and the Brazilian government to understand environmental impacts and development trade-offs.  
*Greenstreet, L., J. Fan, F.S. Pacheco, Y. Bai, M. Eichemberger Ummus, et al. Detecting Aquaculture with Deep Learning in a Low-Data Setting. SigKDD 2023 Fragile Earth Workshop.*  
*F.S. Pacheco, S.A. Heilpern, C. DiLeo, ..., L. Greenstreet, et al. Towards sustainable aquaculture in the Amazon. Nature Sustainability, 2025. doi.org/10.1038/s41893-024-01500-w*
- 2020 - 2021      **Trajectory Inference for Single-Cell Data, University of British Columbia**  
Developed trajectory inference methods to understand spatial and temporal patterns in genomic single-cell sequencing data. Worked closely with medical researchers and biologists to demonstrate the methods in multiple biological systems.

- 2020 - 2021      **Trajectory Inference for Single-Cell Data, University of British Columbia cont.**  
**Greenstreet, L.**, A. Afanassiev, Y. Kijima, M. Heitz, S. Ichiguro, et al. *DNA-GPS: A Theoretical Framework for Optics-Free Spatial Genomics and Synthesis of Current Methods. Cell Systems*, 2023. doi.org/10.1016/j.cels.2023.08.005
- Zhang, S., A. Afanassiev, **L. Greenstreet**, T. Matsumoto, G. Schiebinger. *Optimal transport analysis reveals trajectories in steady-state systems. PLOS Computational Biology*, 2021. doi.org/10.1371/journal.pcbi.1009466
- T.M. Nolan, N. Vukasinovic, C.W. Hsu, J. Zhang, ..., **L. Greenstreet**, et al. *Brassinosteroid gene regulatory networks at cellular resolution in the Arabidopsis root. Science*, 2023. doi.org/10.1126/science.adf4721
- 2019              **Electric Vehicle Infrastructure Planning, University of British Columbia**
- Worked with planners from the City of Surrey, British Columbia to inform their electric vehicle strategy by creating an electric vehicle charger placement model and designing and testing a dashboard with city planners to visualize model outputs alongside socio-economic and planning data.
- Greenstreet, L.**, E.Y. Lai, K. Lin, G.A. Rodriguez-Arelis, R. Ng. *Developing a Smart Electric Vehicle Strategy: From Data to Decisions. ICDE Workshop on Data-Driven Smart Cities*, 2025.
- Greenstreet, L.**, and E.Y. Lai. *Developing a Data-Driven Electric Vehicle Strategy in Surrey, BC. SigKDD Social Impact Session*, 2020.

## FELLOWSHIPS AND AWARDS

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- 2025              **Siegel PiTech PhD Impact Fellowship**, Dept. of Computer Science, Cornell University
- 2022-2026      **Graduate Teaching Award x5**, Dept. of Computer Science, Cornell University
- 2020              **Undergraduate Summer Research Award**, Natural Sciences and Engineering Research Council of Canada (NSERC)
- 2019              **Data Science for Social Good Fellowship**, UBC Data Science Institute

## INTERNSHIPS AND EXPERIENCE

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- 2026              **Emerging Talent Graduate Intern**, Metropolitan Transit Authority
- 2020-2021      **Research Lab Assistant**, Dept. of Mathematics, University of British Columbia

## LEADERSHIP AND SERVICE

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- 2024              **Organizer**, AI for Science Seminar
- 2023              **Organizer**, NeurIPS Computational Sustainability Workshop
- 2023              **Mentor**, BURE Undergraduate Research Program, Cornell
- 2023-2024      **Assistant Organizer**, AI for Science Program, Cornell
- 2020-2021      **Mentor**, Data Science for Social Good Program, UBC Data Science Institute

## TECHNICAL SKILLS

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**Languages:** Python, SQL, R, Julia

**Libraries:** PyTorch, NumPy, Scikit-learn, SciPy, Gurobipy, CVXOPT, Geopandas

**Software:** Git, Anaconda, Linux, Postgres, Azure, Gurobi, PowerBI, Google Cloud, QGIS

**Skills:** deep learning: CNNs, GNNs, transformers, contrastive learning; manifold learning; mathematical programming; statistical modeling; data analysis; data visualization; time-series analysis

## ADDITIONAL PUBLICATIONS

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- 2026 F.S. Pacheco, S.A. Heilpern, R.M. Almeida, ..., **L. Greenstreet**, et al. *Land-use transitions and carbon emissions of aquaculture expansion in the Brazilian Amazon*. Science Advances, 2026. doi.org/10.1126/sciadv.aec5472
- 2025 Li, H., P. Cote, M. Kuoch, J. Ezike, ..., **L. Greenstreet**, et al. *The dynamics of hematopoiesis over the human lifespan*. Nature Methods, 2025. https://doi.org/10.1038/s41592-024-02495-0
- Massri, A.J., A. Berrio, A. Afanassiev, **L. Greenstreet**, et al. *Single-Cell Transcriptomics Reveals Evolutionary Reconfiguration of Embryonic Cell Fate Specification in the Sea Urchin Heliocidaris erythrogramma*. Genome Biology and Evolution, 2025. https://doi.org/10.1093/gbe/evae258
- 2023 Mirka, R., **L. Greenstreet**, M. Grimson, C.P. Gomes. *A New Approach to Finding  $2 \times n$  Partially Spatially Balanced Latin Rectangles*, CP, 2023.
- 2022 Brennan, A., R. Naidoo, **L. Greenstreet**, Z. Mehrabi, N. Ramankutty, C. Kremen. *Functional Connectivity of the Worlds Protected Areas*. Science, 2022. https://doi.org/10.1126/science.abl8974
- Greenstreet, L.**, N.J.A. Harvey, V. Sanches Portella. *Efficient and Optimal Fixed-Time Regret with Two Experts*. ALT, 2022. https://doi.org/10.48550/arXiv.2203.07577
- 2021 Li, H., J. Ezike, A. Afanassiev, **L. Greenstreet**, et al. *Single Cell Analysis Elucidates the Maturation of Human Stem and Progenitor Cell Function from Fetal through Adult Hematopoiesis*. Blood, 2021. https://doi.org/10.1182/blood-2021-151090
- Shahan R., C. Hsu, T.M. Nolan, B.J. Cole, I.W. Taylor, **L. Greenstreet**, et al. *A single cell Arabidopsis root atlas reveals developmental trajectories in wild type and cell identity mutants*. Developmental Cell, 2021. https://doi.org/10.1016/j.devcel.2022.01.008
- 2020 Massri, A.J., **L. Greenstreet**, A. Afanassiev, A. Berrio Escobar, G.M. Wray, G. Schiebinger, D.R. McClay. *Developmental Single-cell transcriptomics in the Lytechinus variegatus Sea Urchin Embryo*. Development, 2020. https://doi.org/10.1242/dev.198614